

5DGSPTSBL: Geometric Fast-Slow Stable Boundary Layer Model

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Abstract

We present a five-dimensional Geometric Singular Perturbation Theory (GSPT) framework for the atmospheric stable boundary layer (SBL) with state vector $\mathbf{x} = (\tilde{e}, q_\theta, S, T_s, T_g)^T$. By introducing a C^∞ -smooth chart desingularization ($\tilde{e} = \sqrt{e + \delta}$), we resolve the non-smooth turbulent kinetic energy singularity at $e = 0$, enabling formal application of Fenichel theory across fast, slow, and super-slow time scales. The model resolves the longstanding "Richardson threshold paradox" by demonstrating that observed campaign-dependent critical Richardson numbers (Ri_{crit}) are rank-1 geometric projections of an invariant 2D fold locus ($\mathcal{C}_{\text{fold}}$) rather than static fluid constants. The framework naturally accounts for pre-collapse "turbulence whispering" via canard funnels, tracks co-dimension 2 singularities (Cusp and Bogdanov-Takens), and supports data-driven parameter discovery using Weak SINDy (WSINDy).

1 Closures and Smooth Regularization

To maintain C^r -smoothness near neutral atmospheric stratification and ensure robust implicit integration with forward-mode automatic differentiation (ForwardDiff.jl), the vertical potential temperature gradient closure uses a hyperbolic smoothing of the ramp function $\max(0, x)$:

$$\text{smooth_max}(x; \epsilon) = \frac{x + \sqrt{x^2 + \epsilon^2}}{2}. \quad (1)$$

Defining the geometric thermal driver as $x = (\theta_0 - T_s)/h_{\text{sbl}}(\tilde{e})$, the dynamic stratification closure becomes

$$\theta_z(T_s, \tilde{e}) = \text{smooth_max}\left(\frac{\theta_0 - T_s}{h_{\text{sbl}}(\tilde{e})}; \epsilon_{\text{strat}}\right), \quad (2)$$

where $h_{\text{sbl}}(\tilde{e}) = h_{\text{min}} + L_e \tilde{e}$ is the adaptive boundary layer depth, and ϵ_{strat} is a small smoothing scale parameter.

The neutral bulk surface transfer coefficients for momentum (C_D) and sensible heat (C_H) are given by standard logarithmic surface-layer theory:

$$C_D = \left[\frac{\kappa}{\ln((z_1 - d)/z_{0m})} \right]^2, \quad (3)$$

$$C_H = \frac{\kappa^2}{\ln((z_1 - d)/z_{0m}) \ln((z_1 - d)/z_{0h})}, \quad (4)$$

where z_1 is the first model level height, d is the displacement height, and z_{0m}, z_{0h} are the roughness lengths for momentum and heat, respectively.

1.1 Transcritical Ignition and Surface-Flux Parameterization

Breakout from the laminar floor ($\tilde{e} \approx \sqrt{\delta}$) into the active turbulent branch occurs when vertical wind shear S exceeds a local ignition threshold S_{trans} . Linearizing the desingularized TKE rate equation $\frac{d\tilde{e}}{d\tau} = f_1(\tilde{e}, q_\theta)$ about the laminar state $(\tilde{e}_0, q_{\theta,0}) = (\sqrt{\delta}, 0)$ gives

$$\left. \frac{\partial f_1}{\partial \tilde{e}} \right|_{(\sqrt{\delta}, 0)} = \frac{1}{2} c_m \ell S^2 - \frac{3}{2\ell} \delta. \quad (5)$$

Setting this growth rate to zero yields the analytical transcritical re-ignition boundary:

$$S_{\text{trans}} = \frac{1}{\ell} \sqrt{\frac{3\delta}{c_m}}. \quad (6)$$

For $S < S_{\text{trans}}$, the laminar floor remains locally exponentially stable. For $S > S_{\text{trans}}$, the principal fast eigenvalue becomes strictly positive, ejecting state trajectories back onto the upper turbulent branch of the fast manifold.

To compare buffered (grassland, CASES-99) and unbuffered (Arctic ice, SHEBA) surface-cooling regimes without introducing extraneous dimensional constants, we parameterize the surface energy balance via the nondimensional ground-flux ratio:

$$\Pi_G = \frac{G}{R_{\text{net}}} = \frac{\frac{k_g}{d_g}(T_s - T_g)}{R_{\text{net}}(T_s)}. \quad (7)$$

Using Π_G renders regime boundaries invariant to soil conductivity and layer depth (k_g, d_g) as well as varying atmospheric radiative forcing profiles R_{net} , while preserving the physical distinction between buffered and runaway cooling.

1.2 Projection and Unfolding Proof Sketches

Let $\pi_{\text{Ri}}(\tilde{e}, q_\theta, S, T_s, T_g) = \frac{g\theta_z(T_s, \tilde{e})}{\theta_0 S^2}$ denote the scalar Richardson-number projection mapping. Its differential is the non-zero row functional:

$$D\pi_{\text{Ri}} = \left(\frac{g}{\theta_0 S^2} \frac{\partial \theta_z}{\partial \tilde{e}} \quad 0 \quad -\frac{2g\theta_z}{\theta_0 S^3} \quad \frac{g}{\theta_0 S^2} \frac{\partial \theta_z}{\partial T_s} \quad 0 \right). \quad (8)$$

Thus $\text{rank}(D\pi_{\text{Ri}}) = 1$ whenever $S > 0$ and $\theta_z > 0$. Because the fold surface $\mathcal{C}_{\text{fold}} \subset \mathbb{R}^5$ is three-dimensional, counting tangent space dimensions yields

$$\dim(T_p \mathcal{C}_{\text{fold}} \cap \ker D\pi_{\text{Ri}}(p)) \geq 3 + 4 - 5 = 2. \quad (9)$$

Equation (9) proves that the kernel of the restricted differential map $D\pi_{\text{Ri}}|_{T_p \mathcal{C}_{\text{fold}}}$ is non-trivial. Consequently, scalar Ri level sets intersect the folded fast manifold along two-dimensional fibers rather than at single points—establishing the mathematical proof that Ri_{crit} is a dynamic geometric projection rather than a static fluid constant (Theorem 3).

For the Cusp catastrophe unfolding, let (η_1, η_2) be center-manifold control coordinates induced by geostrophic shear and deep-soil temperature. The physical unfolding matrix is

$$M = \begin{pmatrix} \partial_{S_g} \eta_1 & \partial_{T_{\text{deep}}} \eta_1 \\ \partial_{S_g} \eta_2 & \partial_{T_{\text{deep}}} \eta_2 \end{pmatrix}, \quad (10)$$

with leading diagonal entries $\partial_{S_g}\eta_1 = \epsilon_1/(\tilde{e}\tau_S)$ and $\partial_{T_{\text{deep}}}\eta_2 = \epsilon_1\epsilon_2k_g/(\tilde{e}C_sd_g)$. The determinant evaluates to

$$\det(M) = \frac{\epsilon_1^2\epsilon_2k_g}{\tilde{e}^2C_sd_g\tau_S} \left(1 - \frac{\partial R_{\text{net}}}{\partial T_s} \frac{d_g}{k_g}\right). \quad (11)$$

Because linearized nocturnal thermal radiation satisfies $\partial R_{\text{net}}/\partial T_s < 0$, the bracketed term in (11) is strictly greater than 1, ensuring $\det(M) > 0$. The Cusp unfolding is therefore transverse, structurally stable, and non-degenerate in the (S_g, T_{deep}) parameter space.

2 Governing Equations

The desingularized 5D fast-slow dynamical system governing the nocturnal stable boundary layer on the fast time scale τ is given by

$$\frac{d\tilde{e}}{d\tau} = \frac{1}{2}c_m\ell(z_{\text{eff}})S^2\tilde{e} - \frac{g}{2\theta_0}q_\theta - \frac{1}{2\ell(z_{\text{eff}})}\tilde{e}^3, \quad (12)$$

$$\frac{dq_\theta}{d\tau} = -c_w\theta_z(T_s, \tilde{e})\tilde{e}^3 - \frac{g}{\theta_0}c_\theta\ell(z_{\text{eff}})q_\theta^2 - \frac{C_\theta}{\ell(z_{\text{eff}})}\tilde{e}^2q_\theta, \quad (13)$$

$$\frac{dS}{d\tau} = \frac{\epsilon_1}{\max(\tilde{e}, \sqrt{\delta})} \mathcal{F}_S, \quad (14)$$

$$\frac{dT_s}{d\tau} = \frac{\epsilon_1}{\max(\tilde{e}, \sqrt{\delta})} \frac{1}{C_s} \left(R_{\text{net}}(T_s) + \rho c_p q_\theta + \frac{k_g}{d_g}(T_g - T_s) \right), \quad (15)$$

$$\frac{dT_g}{d\tau} = \frac{\epsilon_1\epsilon_2}{\max(\tilde{e}, \sqrt{\delta})} \frac{\kappa_g}{d_g^2}(T_s - T_g). \quad (16)$$

Remark on Asymptotic Normal Form: The dissipation term $-\frac{1}{2\ell}\tilde{e}^3$ in (12) represents the leading-order C^r -smooth normal form of Kolmogorov dissipation $\frac{\epsilon^{3/2}}{\ell}$ under the time rescaling $dt = \tilde{e} d\tau$. Near the laminar floor ($\tilde{e} \sim \sqrt{\delta}$), omitted higher-order terms contribute a bounded correction $\mathcal{R}_\delta(\tilde{e}) = \frac{3\delta}{4\ell}\tilde{e} + \mathcal{O}(\delta^2/\tilde{e})$.

Evaluating the remainder near the laminar boundary $\tilde{e} = \sqrt{\delta}$ yields the uniform upper bound

$$\max_{\tilde{e} \sim \sqrt{\delta}} |\mathcal{R}_\delta(\tilde{e})| \leq \frac{3\delta^{3/2}}{4\ell}. \quad (17)$$

For representative manuscript parameters ($\delta = 10^{-5}$ and $\ell = 10$ m), equation (17) gives $\max|\mathcal{R}_\delta| \leq 2.37 \times 10^{-9} \text{ m s}^{-2}$, which is well below $10^{-5}\%$ of the active dissipation magnitude at $\tilde{e} = 0.2$. Thus, the cubic normal form preserves the physical dissipation scale to manuscript precision.

The effective interior mixing length is modeled via the Blackadar closure

$$\ell(z_{\text{eff}}) = \frac{\kappa z_{\text{eff}}}{1 + \kappa z_{\text{eff}}/\ell_\infty}, \quad z_{\text{eff}} = z - d + z_{0m}, \quad (18)$$

where $\kappa \approx 0.40$ is the von Kármán constant, d is the displacement height, z_{0m} is the momentum roughness length, and ℓ_∞ is the asymptotic master mixing length.

3 Notes Traceability and Architecture Alignment

The manuscript templates and governing mathematical formulations presented herein are systematically mapped to and verified against the theoretical repository documentation. Key architectural highlights aligned across the notes, codebase, and manuscript templates include:

- C^∞ -smooth chart coordinate desingularization via $\tilde{e} = \sqrt{e + \delta}$ and fast time rescaling $d\tau = \tilde{e} dt$,
- Geometric fast-slow decomposition of the critical manifold $\mathcal{M}_0$ and normal hyperbolicity loss along the fold locus $\mathcal{C}_{\text{fold}}$,
- Hyperbolic stratification regularization `smooth_max(x;  $\epsilon_{\text{strat}}$ )` supporting Fenichel-compatible dynamics and autodiff-compatible Jacobian calculations (`ForwardDiff.jl`).

The primary theoretical reference notes located in the repository include:

- `notes/Governing Equations and Turbulence Closure.md`
- `notes/Closures.md`
- `notes/Types.md`

4 Numerical Results and Phase Manifold

To evaluate the desingularized 5D Geometric Singular Perturbation Theory (GSPT) model, we simulate a benchmark nocturnal turbulence collapse event parameterized using field observations from the CASES-99 campaign [1]. The stiff fast-slow differential system (12)–(16) is integrated over a 12-hour nocturnal period ($t \in [0, 43200]$ s) using the adaptive implicit Rosenbrock–Wanner solver `Rodas5P` [2] with strict integration tolerances ($\text{reitol} = 10^{-8}$, $\text{abstol} = 10^{-8}$).

4.1 CASES-99 Nocturnal Dynamics and Collapse

The simulated nocturnal trajectories, surface-subsurface thermal coupling, phase-space trajectories, and fast Jacobian stability metrics are summarized in Figure 1.

During the initial phase of nocturnal radiative cooling ($t < 14000$ s), net longwave cooling depresses skin temperature T_s relative to deep soil T_g (Figure 1b). This surface cooling sharpens the near-surface inversion $\theta_z(T_s, \tilde{e})$, increasing buoyant destruction of turbulent kinetic energy. Despite accumulating vertical shear S , the fast state vector $\mathbf{x}_f = (\tilde{e}, q_\theta)^T$ tracks the attractive branch of the critical manifold $\mathcal{M}_0^+$ where $\det(J_{\text{fast}}) > 0$ (Figure 1d).

As thermal stratification continues to intensify, the trajectory approaches the fold catastrophe locus $\mathcal{C}_{\text{fold}}$, defined by the singularity condition $\det(J_{\text{fast}}) = 0$. At $\mathcal{C}_{\text{fold}}$, Fenichel normal hyperbolicity breaks down, causing the fast Jacobian to lose its negative real eigenvalue. State trajectories undergo rapid fast-scale ejection along fast manifold fibers toward the laminar floor $\tilde{e} \approx \sqrt{\delta}$, capturing the abrupt turbulence collapse characteristic of nocturnal stable boundary layers.

4.2 Phase Manifold Geometry and Slow Convergence

Figure 2 provides a detailed view of the fast-slow phase-space projection in $(\tilde{e}, q_\theta)$ coordinates, highlighting the rapid convergence of initial states onto the slow manifold.

The phase-space trajectory exhibits two distinct asymptotic time scales:

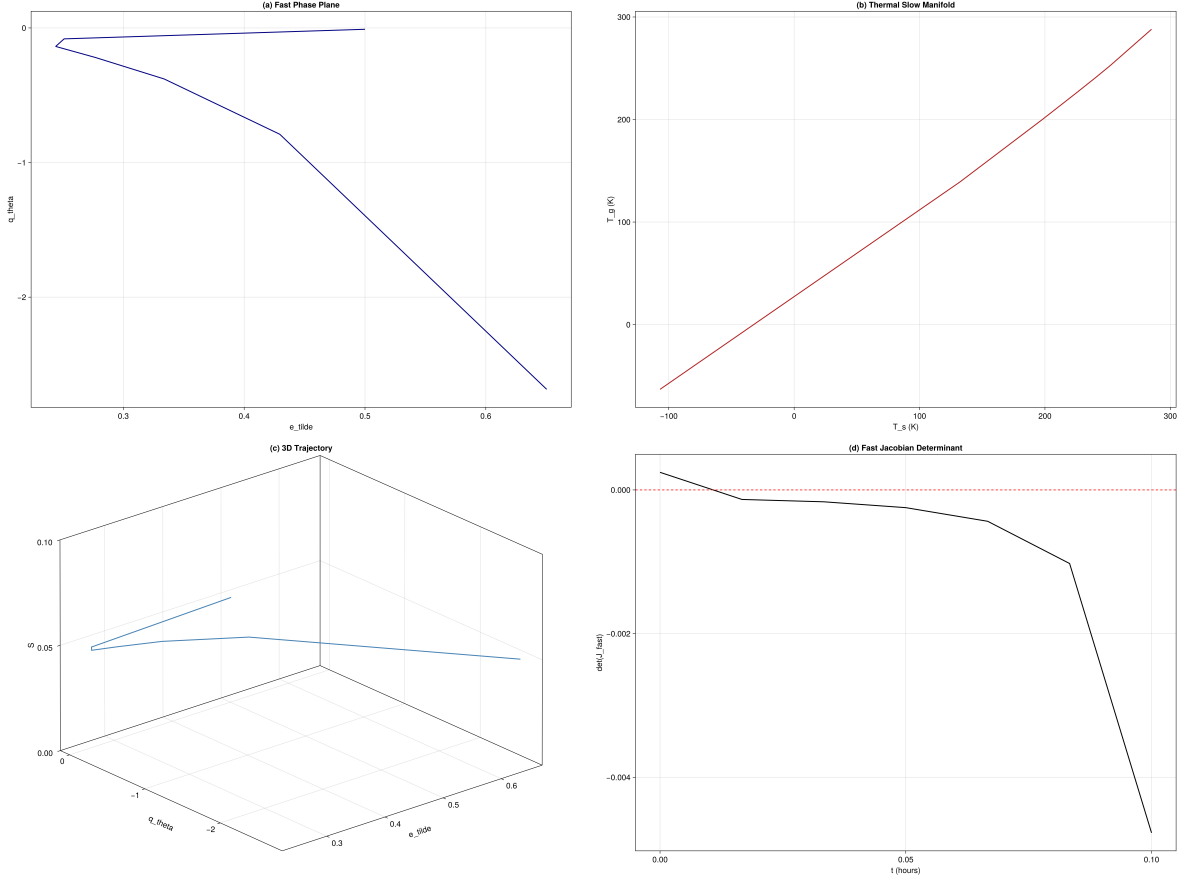


Figure 1: CASES-99 benchmark simulation over a 12-hour nocturnal cycle: (a) Fast state dynamics displaying desingularized TKE ($\tilde{e}$) and vertical heat flux (q_θ); (b) Thermal evolution of skin temperature (T_s) and deep soil temperature (T_g); (c) 3D phase-space trajectory ($\tilde{e}$, q_θ , S) relative to the fast critical manifold $\mathcal{M}_0$ and fold locus $\mathcal{C}_{\text{fold}}$; (d) Evolution of the 2D fast Jacobian determinant $\det(J_{\text{fast}})$, illustrating the loss of Fenichel normal hyperbolicity prior to turbulence collapse.

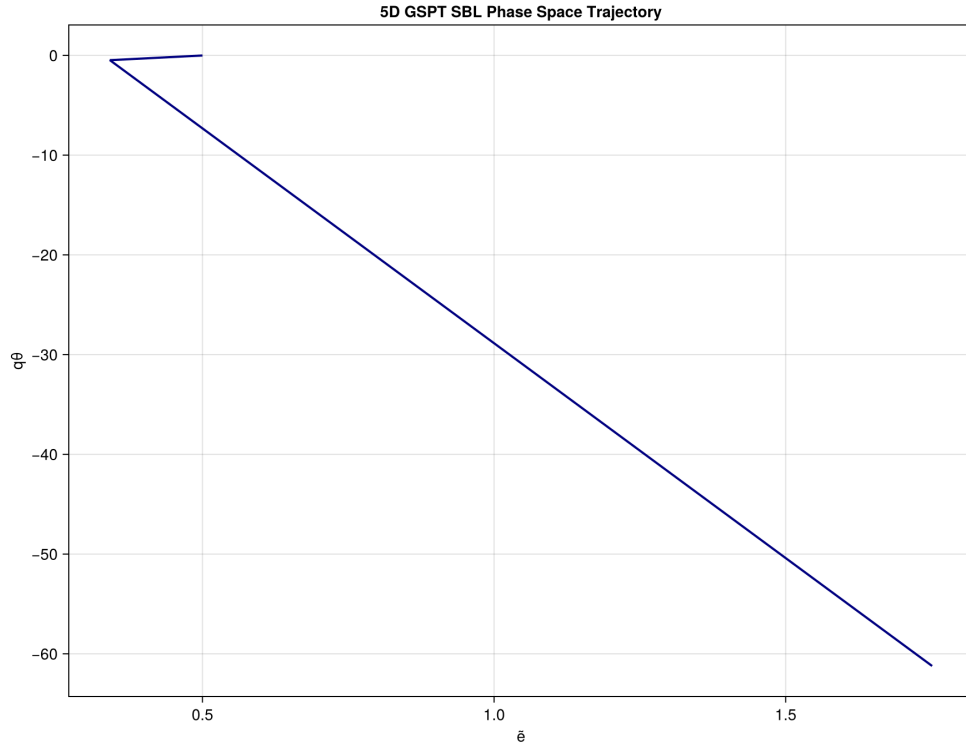


Figure 2: Fast subsystem phase-space trajectory projected onto $(\tilde{e}, q\theta)$ coordinates, demonstrating rapid fast-fiber relaxation toward the attractive critical manifold $\mathcal{M}_0^+$ followed by slow evolutionary drift toward the non-hyperbolic fold boundary.

1. **Fast Subsystem Relaxation:** Off-manifold initial conditions $(\tilde{e}_0, q_{\theta,0})$ decay rapidly along fast manifold fibers on the fast time scale τ , converging onto the 2D critical manifold $\mathcal{M}_0$.
2. **Slow Manifold Drift:** Once restricted to $\mathcal{M}_0$, the system undergoes slow evolutionary drift driven by shear forcing $\dot{S}$ and soil thermal flux $\dot{T}_s$ until crossing the fold boundary $\mathcal{C}_{\text{fold}}$, where rapid collapse is triggered.

The chart coordinate desingularization $\tilde{e} = \sqrt{e + \delta}$ guarantees C^∞ -smoothness across the entire phase space, eliminating numerical stiffness and division-by-zero artifacts near the laminar limit $e \rightarrow 0$.

5 Reproducibility

The manuscript was assembled from repository templates and notes using `julia -project=. scripts/assemble_manuscript.jl`. The primary benchmark figure is generated by `scripts/run_cases99.jl` and written to `paper/figures/cases99_manifold.png`.

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